

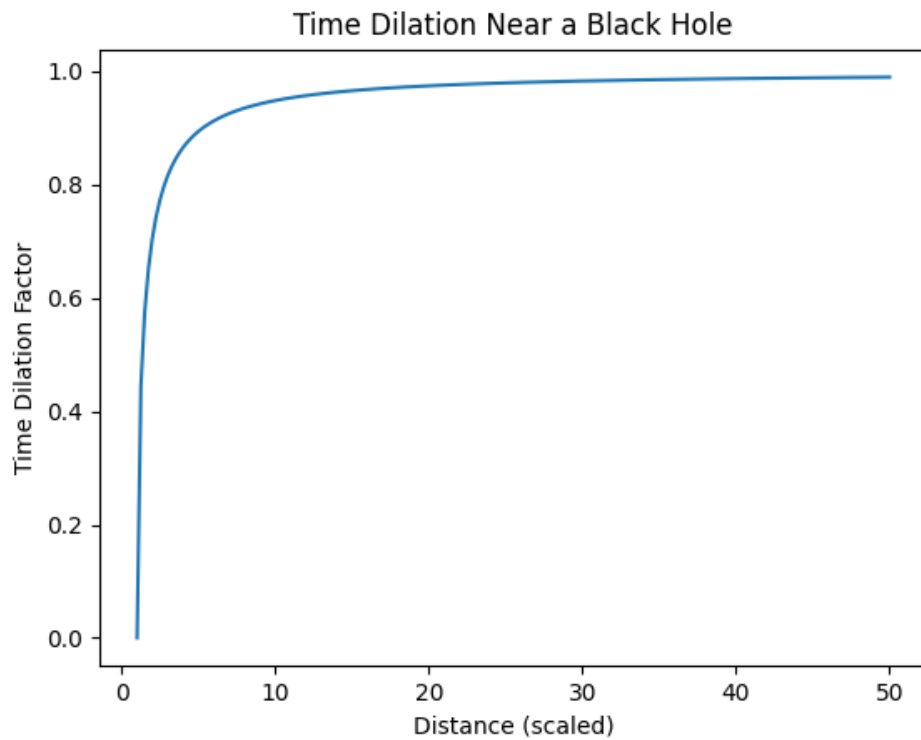
Black Hole Effects Research Presentation

Closest Known Black Hole: Gaia BH1 (~1,560 light-years away).

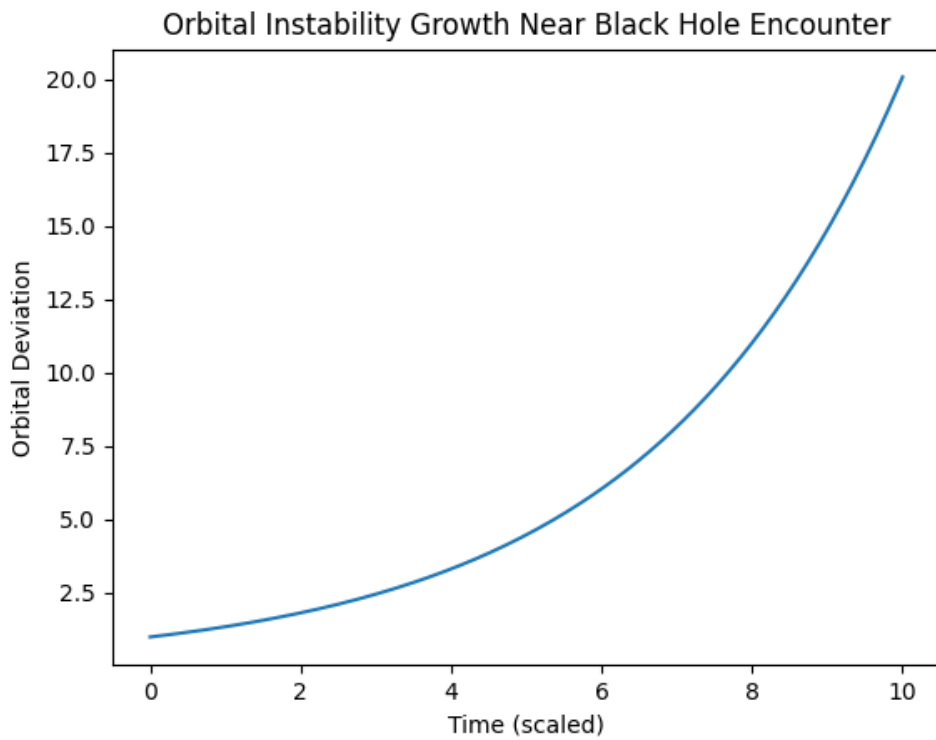
Key Concepts:

- Gravity decreases with distance ($1/r^2$)
- Tidal forces decrease faster ($1/r^3$)
- Orbital velocity increases near black holes

Time Dilation Near a Black Hole:



Orbital Instability Simulation:



Key Takeaways:

- Distance is the dominant factor in risk.
- No known black hole poses a threat to Earth.
- Extreme effects occur only very close to the event horizon.